Mahindra University Hyderabad
École Centrale School of Engineering
Minor II Regular Examination
(Batch 2024)

Program: B. Tech Branch: AI, CSE, ECE, ECM, EEM, VLSI, ME, MT, AE, & NT
Year: I **Semester: I**

Subject: Chemistry - I (CH 1101)

Date: 24/10/2024 **Start Time: 02.00 PM**
Time Duration: 1.5 Hours **Max. Marks: 30**

Instructions

1. Answer all questions.
2. No clarifications will be entertained during the examination.
3. If any information is missing, make appropriate assumptions and proceed.

Table 1: Useful physical constants and conversion factors

Speed of light (c)	=	2.997×10^{8} m s^{-1}
Mass of electron (m_e)	=	9.109×10^{-31} kg
Planck constant (h)	=	6.626×10^{-34} J s
Boltzmann constant (k_B)	=	1.381×10^{-23} J K^{-1}
Avogadro number (N_A)	=	6.023×10^{23} mol^{-1}
Elementary charge (e)	=	1.602×10^{-19} C
Mass of proton (m_p)	=	1.672×10^{-27} kg
Rydberg Constant (R_H)	=	109677 cm^{-1}
1 eV	=	1.602×10^{-19} J
1 E_h	=	27.211 eV

Q1. (Marks: [7+3=10])

(i) For a free particle moving in a 1-dimensional box (of length a) with infinite potential energy walls, calculate the probability of finding the particle in region: $a/3 \leq x \leq 2a/3$, if it is in its ground state (n=1)?

(ii) A particle is moving freely in a ring (of radius r) placed on a plane. State boundary conditions for the particle. Choose the correct form of its wavefunction from the following options:

$$\text{(a) } \psi(\phi) = \sqrt{\tfrac{1}{2\pi}}\, e^{\mathrm{im_l}\phi} \qquad \text{(b) } \psi(\phi) = \tfrac{1}{2\pi}\, e^{\mathrm{im_l}\phi}$$

Q2. (Marks: [5+5=10])

(i) What are the quantum numbers of the wavefunction given below:

$$\Psi_{n,l,m_l}(r,\theta,\phi) = R_{2,1}(r)Y_{1,0}(\theta,\phi)$$

Recognize the atomic orbital (AO) for the above wavefunction. How many radial node and angular node are there for this AO? Using this AO demonstrate linear combination of atomic orbitals.

(ii) Write ground-state electronic configurations of F_2 using molecular orbital (MO) theory (label each MO as either σ or π, and either u or g)? Which one of the following would you expect to have stronger bond: (a) F_2 (b) F_2^+

Q3. (Marks: [5+3+2=10])

(i) A vibrational ground state and an excited state in a molecule are separated by a certain energy gap of ΔE. Given that in a collection of molecules maintained at room temperature (300 K), 0.50% of molecules are already occupying the vibrational excited state. Determine the characteristic peak position (expressed in cm^{-1}) of the spectroscopic transition between these two vibrational energy levels.

(ii) The molar absorption coefficient of an organic molecule at a particular wavelength is 425 L mol^{-1} cm^{-1}. When light of that wavelength passes through a 6.50 mm cell containing a solution of the organic molecule, 48.3% of the light is absorbed. Determine the concentration of this solution in mol/L.

(iii) Arrange the following electronic transitions in the increasing order of their wavelengths (λ) and designate each of them as either strong or weak peak:

(a) $n \rightarrow \pi^*$ (b) $\sigma \rightarrow \sigma^*$ (c) $\pi \rightarrow \pi^*$